NO COOL AIR [FULL-AUTO AIR CONDITIONER]

id0703c1700900

DESCRIPTION	• Magnetic clutch does not operate
POSSIBLE CAUSE	• Airflow mode system malfunction • Blower system malfunction • A/C compressor malfunction • CAN communication malfunction • PCM A/C cut control system malfunction • Refrigerant pressure sensor malfunction • Evaporator temperature sensor malfunction • Magnetic clutch malfunction • A/C MAG 7.5 A fuse malfunction • A/C relay malfunction • Improper refrigerant charging amount

- When performing an asterisked (*) troubleshooting inspection, shake the wiring harness and connectors while performing the inspection to discover whether poor contact points are the cause of any intermittent malfunctions. If there is a problem, check to make sure connectors, terminals and wiring harnesses are connected correctly and undamaged.

Diagnostic Procedure

Step	Inspection		Action
1	**CHECK MALFUNCTION SYMPTOMS** **Note** • Without i-stop, go to the next step. • Is malfunctions occur in only when operating the i-stop?	Yes	Perform the i-stop troubleshooting. (See FOREWORD [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See FOREWORD [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**INSPECT AIR BLOW OUT** • Does air blow out?	Yes	Go to the next step.
		No	Go to Step 1 of the following troubleshooting. (See INSUFFICIENT AIR (OR NO AIR) BLOWN FROM VENTS [FULL-AUTO AIR CONDITIONER].) (See AMOUNT OF AIR BLOWN FROM VENTS DOES NOT CHANGE [FULL-AUTO AIR CONDITIONER].)
3	**INSPECT A/C COMPRESSOR OPERATION** • Start engine. • Turn A/C switch and fan switch on. • Does A/C compressor operate?	Yes	Go to Step 1 of the following troubleshooting. AIR FROM VENTS NOT COLD ENOUGH [FULL-AUTO AIR CONDITIONER]
		No	Go to the next step.
4	**INSPECT FOR DTCs IN PCM, CLIMATE CONTROL UNIT, AND INSTRUMENT CLUSTER** • Inspect for DTCs related to the PCM, climate control unit, and instrument cluster on-board diagnostic system. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) (See DTC INSPECTION [INSTRUMENT CLUSTER].) • Are any DTCs displayed?	Yes	Go to appropriate inspection procedure. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].) (See DTC TABLE [FULL-AUTO AIR CONDITIONER].) (See DTC TABLE [INSTRUMENT CLUSTER].)
		No	Go to the next step.

Step	Inspection		Action
5	**DETERMINE IF MALFUNCTION CAUSE IS A/C REQUEST SIGNAL OR A/C RELAY OPERATIONAL MALFUNCTION** • Access PCM PID ACCS using the M-MDS. (See OBD PARAMETER IDENTIFICATION (PID) ACCESS [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See OBD PARAMETER IDENTIFICATION (PID) ACCESS [SKYACTIV-D 1.5].) • Start the engine and idle it. • Turn the PID ACCS to ON from OFF using the M-MDS simulation function. • Is the A/C magnetic clutch engaged?	Yes	Go to the next step.
		No	Go to Step 10.
6	**DETERMINE IF MALFUNCTION CAUSE IS A/C PRESSURE SENSOR SIGNAL OR ELSEWHERE** • Access climate control unit PID AC_PRES. (See OBD PARAMETER IDENTIFICATION (PID) ACCESS [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See OBD PARAMETER IDENTIFICATION (PID) ACCESS [SKYACTIV-D 1.5].) • Monitor the AC_PRES PID while turning on and off the air conditioner by switching the control panel. • Is the PID normal?	Yes	Go to Step 8.
		No	Go to the next step.
7	**INSPECT REFRIGERANT PRESSURE SENSOR** • Inspect the refrigerant pressure sensor. (See REFRIGERANT PRESSURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is the refrigerant pressure sensor normal?	Yes	Inspect the following and repair or replace if necessary. • Refrigerant charging amount • A/C compressor for seizure Then go to Step 16.
		No	Repair or replace malfunctioning part according to inspection result, then go to Step 16.
8	**DETERMINE IF MALFUNCTION CAUSE IS EVAPORATOR TEMPERATURE SENSOR OPERATIONAL SIGNAL OR ELSEWHERE** • Access climate control unit PID EVA_TMP_SEN. (See PID/DATA MONITOR DISPLAY [FULL-AUTO AIR CONDITIONER].) • Monitor the EVA_TMP_SEN PID while turning on and off the air conditioner by switching the control panel. (See PID/DATA MONITOR TABLE [FULL-AUTO AIR CONDITIONER].) • Is the PID normal?	Yes	Go to Step 10.
		No	Go to the next step.
9	**INSPECT EVAPORATOR TEMPERATURE SENSOR** • Inspect the evaporator temperature sensor. (See EVAPORATOR TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is the evaporator temperature sensor normal?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between evaporator temperature sensor terminal and climate control unit terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 16.
		No	Replace the evaporator temperature sensor, then go to Step 16. (See A/C UNIT DISASSEMBLY/ASSEMBLY.)

Step	Inspection		Action
10*	**INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY) IS IN A/C CONTROL SIGNAL CIRCUIT (BETWEEN A/C RELAY AND PCM) OR ELSEWHERE** • Does cool air blow out when terminal E of A/C relay connector (A/C control signal) is grounded?	Yes	Release short, then go to the next step.
		No	Go to Step 12.
11*	**INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY) IS IN PCM OR WIRING HARNESS (BETWEEN A/C RELAY AND PCM)** • Test voltage at the A/C relay control signal terminal of PCM. • Is voltage **approx. 12 V**?	Yes	Inspect the PCM, then go to Step 16. (See PCM INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See PCM INSPECTION [SKYACTIV-D 1.5].)
		No	Refer to the wiring diagram and verify whether or not there is a common connector between A/C relay terminal and PCM terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 15.
12	**INSPECT TO SEE WHETHER MALFUNCTION IS IN MAGNETIC CLUTCH OR ELSEWHERE** • Inspect the magnetic clutch. (See MAGNETIC CLUTCH INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See MAGNETIC CLUTCH INSPECTION [SKYACTIV-D 1.5].) • Is the magnetic clutch normal?	Yes	Go to the next step.
		No	Replace the magnetic clutch, then go to Step 16. (See MAGNETIC CLUTCH DISASSEMBLY/ASSEMBLY [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See MAGNETIC CLUTCH DISASSEMBLY/ASSEMBLY [SKYACTIV-D 1.5].)
13	**INSPECT FUSE** • Remove the A/C MAG 7.5 A fuse. • Is the fuse normal?	Yes	Go to the next step.
		No	Replace the A/C MAG 7.5 A fuse, then go to Step 16. If fuse burns out immediately, go to the next step.
14	**INSPECT WIRING HARNESS BETWEEN FUSE BLOCK AND A/C RELAY FOR LACK OF CONTINUITY** • Test voltages at following terminals of A/C relay. — Terminal A (A/C relay control signal) — Terminal C (A/C control signal) • Are voltages **approx. 12 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between fuse block terminal and A/C relay terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 16.
15	**INSPECT TO SEE WHETHER MALFUNCTION IS IN A/C RELAY OR WIRING HARNESS (BETWEEN A/C RELAY AND MAGNETIC CLUTCH)** • Test voltage at the following terminal of A/C relay. — Terminal D (magnetic clutch operation signal) • Is voltage **approx. 12 V**?	Yes	Inspect wiring harness between A/C relay and magnetic clutch. • If above wiring harness is normal, go to the next step. • If above wiring harness malfunctions, repair wiring harness, then go to the next step.
		No	Replace the A/C relay, then go to the next step. (See RELAY LOCATION.)
16	**VERIFY THAT MALFUNCTION SYMPTOMS DO NOT RECUR AFTER REPAIR** • Does cool air discharge from the vents? (Are the results of refrigerant system performance test normal?) (See REFRIGERANT SYSTEM PERFORMANCE TEST.)	Yes	Troubleshooting completed. Explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.